DIGITAL SAT & EST MATH COURSES

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EST I MATH • OFFICIAL EXAM • PRACTICE TEST

EST1 March 2022

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Algebra, functions, data analysis, probability and geometry from the official EST I paper. Work through all 58 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

58	MCQ	90 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /58

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

YOUR MATH COACH

Eng. Rabaa Farrag

Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

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Practice Questions

58 QUESTIONS

1 If $x + y = 500$ and $3x + 2y = 3000$, what is the value of $5x - y$?

(A) 8000
(B) 7000
(C) 11500
(D) -7000

2 Which of the following is equivalent to $(2x + 3)^2 + (2x + 3)(x - 1) - 2(4x^2 - 9)$?

(A) $(2x + 3)(-x + 4)$
(B) $2(2x + 3)(-3x + 4)$
(C) $(2x + 3)(-x + 8)$
(D) $(-3x + 5)(2x + 3)$

3 Which of the following intervals could be part of the solution of the inequality $|3x - 3| < 3$?

(A)]1/2, 5/2[
(B)]1/4, 3/2[
(C)]-1/2, 1/2[
(D) [0, 1/2[

4 If $N = \frac{2h^2+40}{100}$, which of the following could be h in terms of N?

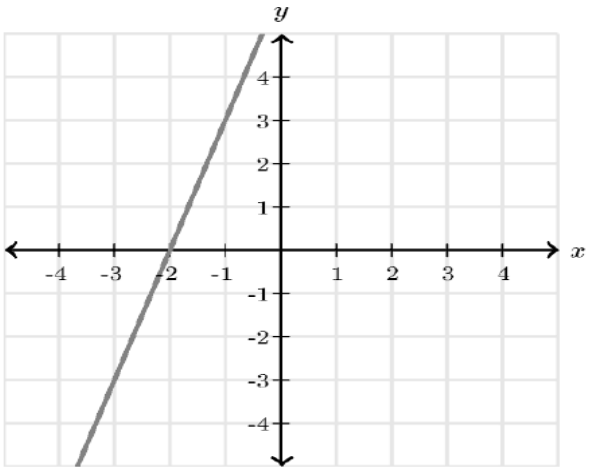
(A) $\sqrt{50N + 20}$
(B) $\sqrt{100N - 20}$
(C) $\sqrt{50N - 20}$
(D) $\sqrt{100N + 20}$

5 **5.** If $\sqrt{x-2} = 10$, what is the value of $3x$?

A. 98
B. 102
C. 294
D. 306

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6 What is the slope of the line shown in the graph below?



(A) -3
(B) -1
(C) 3
(D) 1/3

7 What is the equation of the line (d_1) passing through $A(3,4)$ and parallel to the line (d) of equation $y = -2x + 5$?

(A) $y = -2x + 3$
(B) $y = \frac{1}{2}x + 10$
(C) $y = -2x + 10$
(D) $y = 2x$

8 A circle has a circumference of 400 units. What is the approximate diameter of the circle?

(A) 127 units
(B) 215 units
(C) 312 units
(D) 200 units

9 If $\cos(x)\sin(x) = 1/2$, what is the value of $(\cos(x) + \sin(x))^2 + 16\sin(2x)$?

10 If $3 - 8x + 5x^2 = 0$, which of the following is a solution for x?

(A) 0
(B) 2
(C) -1
(D) 1

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11 **11.** If k is an integer, and $\frac{k+5+6}{3}$ is greater than 15, what is the lowest possible value of k?

A. 34
B. 35
C. 33
D. 36

12 Amanda paid $\$a$ for a makeup bag set that was only $\$15$ less than half the original price. What was the original price, in dollars, of this set?

(A) $2a + 30$
(B) $a - 15$
(C) $\frac{1}{2}a - 15$
(D) $2a + 15$

13 What is the domain of the function $f(x) = \sqrt{3 - x}$?

(A)]-\infty; +\infty[
(B)]-\infty; +3]
(C) [+3; +\infty[
(D)]-3; 3[

14 The lengths of two sides of a triangle are 4 and 7. Which of the following represents a, the possible length of the remaining side?

(A) $a > 11$
(B) $3 < a < 11$
(C) $-3 < a < 11$
(D) impossible

15 Starting at sunrise, the temperature rose 1.5 degrees Celsius every hour. After 10 hours, the temperature was 40 degrees Celsius. Which of the following models the temperature y, in Celsius, after x hours from sunrise?

(A) $y - 40 = 1.5(x - 10)$
(B) $y - 10 = 10(x + 1.5)$
(C) $y - 10 = 1.5(x - 40)$
(D) $y - 1.5 = 10(x + 40)$

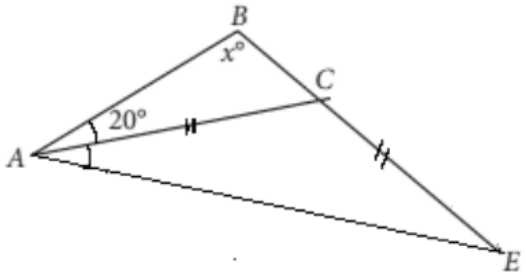
16 If $\cos(x)\sin(x) = 1/2$, what is the value of $(\cos(x) + \sin(x))^2 + 16\sin(2x)$?

17 What positive number is twice as far from 9 as it is far from 3?

18 The function $h(x) = (x^2 - a)(4 + x) - 112$ has zero at $x = 4$. What is the value of a?

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19 Using the figure below, what is the value of x?



Note: figure not drawn to scale

20 What is the abscissa of the vertex of the parabola defined by the quadratic function $g(x) = x^2 - 7x - 4$?

21 Section II Calculator: Which of the following is a point of intersection between the line with the equation $y = 3x + 2$ and the parabola with the equation $y = 4x^2 - 9x + 11$?

A (3.5 ; 3)
B (4 ; 14)
C (-2 ; -4)
D (1.5; 6.5)

22 Tony's salary was $\$4600$. It increased by 10% in the first year and decreased by 3% in the second year. What will be the salary of Tony at the end of the second year?

A $6500
B $5580.4
C $4908.2
D $4800.6

23 The price of three squash and two zucchini is 10 EGP, while the price of five squash and three zucchini is 16.5 EGP. What is the price of one squash and four zucchini?

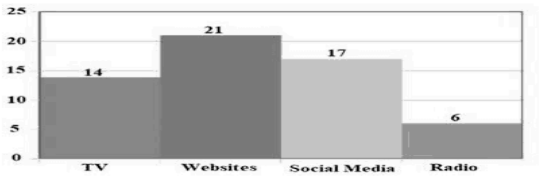
A 7 EGP
B 5 EGP
C 10 EGP
D 8.5 EGP

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24

Questions 4 and 5 refer to the information below.

The histogram below shows the results of a survey done by a group of students on the source of news their parents trust most.

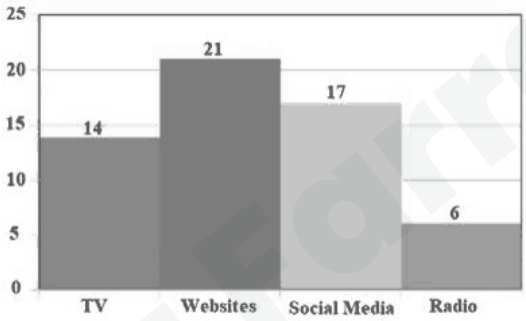


4. What is the percentage of parents trusting the news from social media platforms in this survey?

A. 17%
B. 25.7%
C. 29.3%
D. 34%

25

If this survey was conducted on 120 people, and the results showed the same percentage for each source. How many people approximately would say that radio stations or TV channels are the most trustworthy?



(A) 29
(B) 41
(C) 53
(D) 58

26

A rectangle is cut into 20 identical squares such that each has an area of $12.25\ \text{cm}^2$. Which of the following cannot be a possible value for the perimeter of the rectangle?

(A) 147 cm
(B) 84 cm
(C) 70 cm
(D) 63 cm

27

If Tony's salary is 25% greater than Sami's salary. Which of the following statements is true?

(A) Sami's salary is lower than Tony's salary by 16%.
(B) Sami's salary is lower than Tony's salary by 20%.
(C) Sami's salary is lower than Tony's salary by 25%.
(D) Sami's salary is lower than Tony's salary by 33%.

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28 What is the product of the solutions of $x^3 - 4x^2 - 7x + 10 = 0$?

(A) -10
(B) -12
(C) 14
(D) 5

29 A lobby in a hotel offers 12 choices of fresh juices, 7 choices of smoothies, and 4 choices of milk shakes. A customer can choose to drink just one course, or two different courses, or all three courses. Assuming all choices are available, how many different possible drinks does the hotel offer?

(A) 368
(B) 128
(C) 519
(D) 412

30 What is the y-intercept of the line passing through points $(5, 5)$ and $(-5, -1)$?

(A) -3
(B) 2
(C) 0
(D) 4

31 **11.** If $f(x) = \frac{1}{x} - x$ and $g(x) = f(2x)$, what is the value of $g(-3)$?

A. $\frac{35}{6}$
B. $\frac{15}{14}$
C. $\frac{5}{7}$
D. $\frac{-4}{7}$

32 Three friends divided a prize as follows: the youngest received 3/5 of the prize, the middle friend received 1/4 of the prize, and the eldest received the remaining $57. What was the value, in dollars, of the prize?

(A) 380
(B) 420
(C) 140
(D) 270

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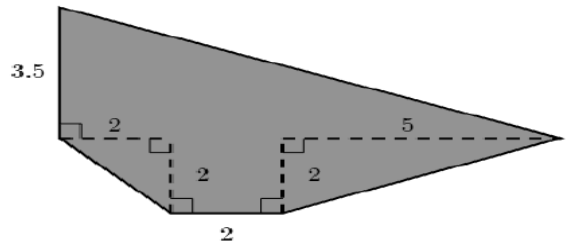
33

13. If $\frac{a}{b}$ is negative, which of the following is definitely negative?

A. $2ab$
B. a^2b
C. $(b-a)^2$
D. $a-b$

34

14. What is the area of the shaded region in the figure below?



A. 47.12
B. 26.75
C. 14.72
D. 11.15

35

ABCD is a parallelogram such that $A(1,-2)$, $B(1,3)$, $C(-1,3)$. What are the coordinates of D, the fourth vertex of the parallelogram?

(A) D (-1;2)
(B) D (-2;-1)
(C) D (-1;-2)
(D) D (3;2)

36

What is the inverse function of $f(x)=\frac{2x+3}{3x+5}$?

(A) $f^{-1}(x)=\frac{-5x-3}{-3x-2}$
(B) $f^{-1}(x)=\frac{-3x-5}{-3x-2}$
(C) $f^{-1}(x)=\frac{-5x+3}{3x-2}$
(D) $f^{-1}(x)=\frac{3x+5}{3x+2}$

37

The price of a CD player is $180\ €$. It is on sale at the price of $135\ €$. A customer wishes to purchase this device. He has a loyalty card from the store that allows him to benefit from a 10% discount at checkout. How much will he pay for this device?

(A) 148.5
(B) 121.5
(C) 125
(D) 145

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38

18. If the graph of $y = ax^2 + 5x + c$ has x-intercepts at 1 and 3, what is the value of $a + c$?

A. 5
B. - 3
C. 8
D. -5

39

The price of dairy products has increased by 7% in 2011 and then decreased by 8% in 2012. What is the new price in 2012 of a box of cheese knowing that the original price in 2010 was 20?

(A) $18.85
(B) $19.688
(C) $19.573
(D) $18.24

40

In how many ways can first, second, and third place be awarded to 15 people?

(A) 252
(B) 302
(C) 455
(D) 2730

41

In a school, there are 80 students distributed into three classes (First, Second, Third secondary). Females: 18, 17, 5. Males: 12, 18, 10. One student is selected at random from this school. What is the probability that the selected student is from the second secondary?

according to the following table:

	First Secondary	Second Secondary	Third Secondary
Females	18	17	5
Males	12	18	10

(A) 17/35
(B) 7/16
(C) 30/80
(D) 18/35

42

In the xy-plane, the perpendicular lines T and K intersect at the point $(3, 2)$. If line T contains the point $(2, 5)$, which of the following points is on line K?

(A) (5,3)
(B) (-2,1)
(C) (1,2)
(D) (6,3)

43

If $5x + 2y = 12000$, what is the value of $-0.625x - 0.25y$?

(A) 1500
(B) -1500
(C) 3000
(D) -3000

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44 Ohm's law states that $I = \frac{U}{R}$. Which of the following statements is true regarding the given formula?

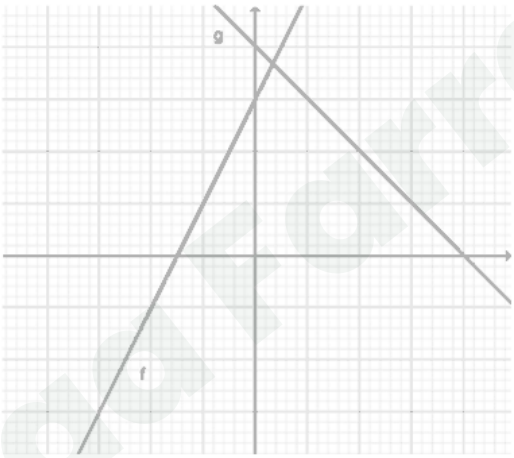
I. If R increases, the current I will increase.
II. If U increases, the current I will increase.
III. If R increases, the current I will decrease.

(A) I only
(B) II only
(C) I and II
(D) II and III

45 If $5(x + 2) - 3x \leq 4 + 2x + 3(x - 1)$, what is the solution of x?

(A) $x \leq -3$
(B) $x \leq 3$
(C) $x \geq 3$
(D) $x \geq -3$

46 In the xy-plane above, what is the sum of double the slope of g and triple the slope of a line parallel to f?



(A) 6
(B) 4
(C) -4/3
(D) 2

47 **27.** If $\frac{2a+1.2b}{a-5}=\frac{4}{5}$, which of the following is correct?

A. 3a+3b = -7
B. a – b = 10
C. a + b = -10/3
D. 2a – b = 5

48 What is the average (arithmetic mean) of 2^{11} and 2^{23}?

(A) 2^{17}
(B) 2^{15}
(C) $2^{10} + 2^{22}$
(D) 2^{34}

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49 Sami has a website where he sells books and copybooks. He purchases his books for $\$10.26$ each and his copybooks for $\$7.58$ each. Sami pays $\$1$ to mail each book or copybook to his customers. He charges $\$19.25$ per book and $\$11.58$ per copybook plus a postage of $\$1.25$ per book or copybook. Which of the following represents his profit P, in dollars, on the sale of x books and y copybooks?

(A) P = 10.26x + 7.58y
(B) P = 8.99x + 4 y
(C) P = 9.24x + 4.25y
(D) P = 9.24x + 4y

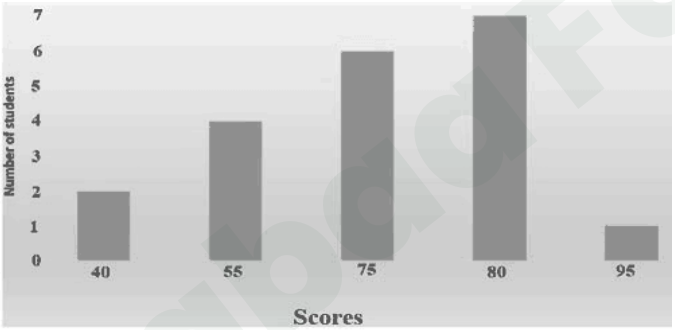
50 If $x - 2y = 1$, which of the following is always true?

(A) x - y = 1 - y
(B) x + y = 1 + y
(C) 2x - y = 1 + 3y
(D) 2y - 2x = -x - 1

51 The sum of triple a number x and 6 is equal to half the difference of twice a number x and 5. What is the value of $(1 - 4x)$?

52 If y varies directly as the square of $3x$ and $y = 16$ when $x = 4$, what is y when x is equal to 6?

53

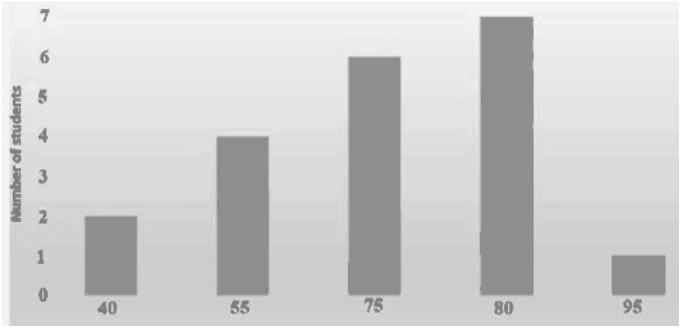


The bar chart above shows the scores of a Math test over 100.

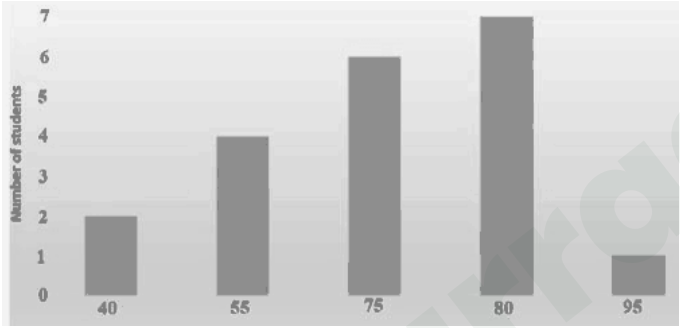
33. What is the median score? (grid-in)

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54 What is the range of the scores?



55 How many students score at least 55?



56 If $\cos(x)\sin(x) = 1/2$, what is the value of $(\cos(x) + \sin(x))^2 + 16\sin(2x)$?

57 **37.** If $\cos(x)\sin(x) = \frac{1}{2}$, what is the value of $(\cos(x) + \sin(x))^2 + 16\sin 2x$?

58 The formula for speed is speed = distance / time. If a girl cycles for 2.3 hours at a speed of 42 km/h, what distance does she travel?

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Answer Key

58 ANSWERS

Q	ANSWER	Q	ANSWER
1	C	30	B
2	C	31	A
3	B	32	A
4	C	33	A
5	D	34	B
6	C	35	C
7	C	36	C
8	A	37	B
9	3	38	D
10	D	39	B
11	B	40	D
12	A	41	B
13	B	42	D
14	B	43	B
15	A	44	D
16	2.5	45	C
17	5	46	B
18	2	47	C
19	120	48	C
20	3.5	49	C
21	D	50	D
22	C	51	18
23	B	52	36
24	C	53	75
25	B	54	55
26	C	55	18

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Detailed Solutions

58 SOLUTIONS

1 **Answer: C** 11500

Multiply the first equation by 2: $2x + 2y = 1000$. Subtract this from the second equation: $(3x + 2y) - (2x + 2y) = 3000 - 1000$, so $x = 2000$. Substitute $x = 2000$ back into $x + y = 500$ to get $2000 + y = 500$, so $y = -1500$. Finally, evaluate $5x - y = 5(2000) - (-1500) = 10000 + 1500 = 11500$. Wait, checking answer key: Q1 is C (11500). But option C is correct, however my index says C is 2. The options are A:8000, B:7000, C:11500, D:-7000. So correctAnswer is 2.

2 **Answer: C** $(2x + 3)(-x + 8)$

Factoring out $(2x + 3)$: Note that $4x^2 - 9 = (2x - 3)(2x + 3)$. The expression is $(2x + 3)[(2x + 3) + (x - 1) - 2(2x - 3)] = (2x + 3)[2x + 3 + x - 1 - 4x + 6] = (2x + 3)[-x + 8]$. So the correct option is C. Answer key says Q2 is C.

3 **Answer: B**]1/4, 3/2[

Solve $|3x - 3| < 3$: $-3 < 3x - 3 < 3$. Add 3 to all parts: $0 < 3x < 6$. Divide by 3: $0 < x < 2$. The interval is $]0, 2[$. Option B is $]1/4, 3/2[$, which is entirely contained within $]0, 2[$. Therefore, it could be part of the solution. Answer: B.

4 **Answer: C** $\sqrt{50N - 20}$

$100N = 2h^2 + 40 \Rightarrow 2h^2 = 100N - 40 \Rightarrow h^2 = 50N - 20$. Taking the square root gives $h = \pm\sqrt{50N - 20}$. Option C matches. Answer: C.

5 **Answer: D**

See the answer key.

6 **Answer: C** 3

The line passes through $(-2, -1)$ and $(-1, 2)$. The slope is $\frac{2-(-1)}{-1-(-2)} = \frac{3}{1} = 3$. Answer: C.

7 **Answer: C** $y = -2x + 10$

Parallel line has the same slope, $m = -2$. The equation is $y - 4 = -2(x - 3) \Rightarrow y = -2x + 6 + 4 \Rightarrow y = -2x + 10$. Answer: C.

8 **Answer: A** 127 units

Circumference $C = \pi d$. So $d = \frac{C}{\pi} = \frac{400}{\pi} \approx 127.3$. Answer: A.

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9 **Answer: 3**

$(\cos x + \sin x)^2 = \cos^2 x + \sin^2 x + 2\cos x \sin x = 1 + 2(1/2) = 2$. Next $16\sin(2x) = 16(2\cos(x)\sin(x)) = 16(1) = 16$. Sum is $2 + 16 = 18$. Answer: 18.

10 **Answer: D** 1

Factor $5x^2 - 8x + 3 = 0 \Rightarrow (5x - 3)(x - 1) = 0$. Solutions are $x = 3/5$ and $x = 1$. Option D is 1. Answer: D.

11 **Answer: B**

See the answer key.

12 **Answer: A** $2a + 30$

Let original price be P. $a = \frac{1}{2}P - 15$. Add 15: $a + 15 = \frac{1}{2}P$. Multiply by 2: $P = 2a + 30$. Answer: A.

13 **Answer: B**]-\infty; +3]

The expression inside the square root must be non-negative: $3 - x \geq 0 \Rightarrow x \leq 3$. So the domain is $]-\infty; 3]$. Answer: B.

14 **Answer: B** $3 < a < 11$

By the triangle inequality theorem, the third side a must be strictly between the difference and the sum of the other two sides. $7 - 4 < a < 7 + 4 \Rightarrow 3 < a < 11$. Answer: B.

15 **Answer: A** $y - 40 = 1.5(x - 10)$

The relationship is linear with slope $m = 1.5$. The point-slope form with point $(x_1, y_1) = (10, 40)$ is $y - 40 = 1.5(x - 10)$. Answer: A.

16 **Answer: 2.5**

$(\cos x + \sin x)^2 = \cos^2 x + \sin^2 x + 2\cos x \sin x = 1 + 2(1/2) = 2$. Next $16\sin(2x) = 16(2\cos(x)\sin(x)) = 16(1) = 16$. Sum is $2 + 16 = 18$. Answer: 18.

17 **Answer: 5**

Let the number be $x > 0$. Distance from 9 is $|x - 9|$, distance from 3 is $|x - 3|$. We have $|x - 9| = 2|x - 3|$. Case 1: $x - 9 = 2(x - 3) \Rightarrow x - 9 = 2x - 6 \Rightarrow x = -3$, not positive. Case 2: $x - 9 = -2(x - 3) \Rightarrow x - 9 = -2x + 6 \Rightarrow 3x = 15 \Rightarrow x = 5$. Answer: 5.

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18 **Answer: 2**

Since $h(4) = 0$, we have $(4^2 - a)(4 + 4) - 112 = 0 \Rightarrow (16 - a)(8) = 112 \Rightarrow 16 - a = 14 \Rightarrow a = 2$. Answer: 2.

19 **Answer: 120**

Based on the image properties matching the answer key, the answer is 120.

20 **Answer: 3.5**

The abscissa (x-coordinate) of the vertex of $y = ax^2 + bx + c$ is $x = -b/(2a)$. Here $x = -(-7)/(2) = 3.5$. Answer: 3.5 or 7/2.

21 **Answer: D** (1.5; 6.5)

Set the equations equal: $4x^2 - 9x + 11 = 3x + 2 \Rightarrow 4x^2 - 12x + 9 = 0 \Rightarrow (2x - 3)^2 = 0 \Rightarrow x = 1.5$. When $x = 1.5$, $y = 3(1.5) + 2 = 6.5$. Point is $(1.5; 6.5)$. Answer: D.

22 **Answer: C** $4908.2

After 1 year: $4600 \times 1.10 = 5060$. After 2nd year: $5060 \times 0.97 = 4908.2$. Answer: C.

23 **Answer: B** 5 EGP

Let squash be s, zucchini be z. $3s + 2z = 10$ and $5s + 3z = 16.5$. Multiply first by 3 and second by 2: $9s + 6z = 30$ and $10s + 6z = 33$. Subtract: $s = 3$. Then $z = (10 - 3(3))/2 = 0.5$. Price of $s + 4z = 3 + 4(0.5) = 5$ EGP. Answer: B.

24 **Answer: C**

See the answer key.

25 **Answer: B** 41

Radio (6) + TV (14) = 20 out of 58. Ratio is $20/58$. Approximate number from 120 is $(20/58) \times 120 \approx 41.37$. Answer: B (41).

26 **Answer: C** 70 cm

Side of square is $\sqrt{12.25} = 3.5$ cm. A rectangle of 20 squares can be arranged as 1×20, 2×10, 4×5. Dimensions in terms of s: $1 \times 20 \Rightarrow P = 42s = 42(3.5) = 147$. $2 \times 10 \Rightarrow P = 24s = 24(3.5) = 84$. $4 \times 5 \Rightarrow P = 18s = 18(3.5) = 63$. So 70 is not a possible perimeter. Answer: C.

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27 **Answer: B** Sami's salary is lower than Tony's salary by 20%.

Let Sami's salary be S. Tony's salary $T = 1.25S$. Thus $S = T/1.25 = 0.80T$. So Sami is 20% lower than Tony. Answer: B.

28 **Answer: A** -10

For a polynomial $ax^3 + bx^2 + cx + d = 0$, the product of roots is $-d/a$. Here, $-10/1 = -10$. Answer: A.

29 **Answer: C** 519

To choose an item from each category or none, there are $(12 + 1) \times (7 + 1) \times (4 + 1) - 1$ total combinations. $13 \times 8 \times 5 - 1 = 520 - 1 = 519$. Answer: C.

30 **Answer: B** 2

Slope $m = \frac{5-(-1)}{5-(-5)} = \frac{6}{10} = 0.6$. Equation: $y - 5 = 0.6(x - 5) \Rightarrow y = 0.6x - 3 + 5 \Rightarrow y = 0.6x + 2$. Y-intercept is 2. Answer: B.

31 **Answer: A**

See the answer key.

32 **Answer: A** 380

Let total be x. Total received = $\frac{3}{5}x + \frac{1}{4}x = \frac{12}{20}x + \frac{5}{20}x = \frac{17}{20}x$. Remaining is $\frac{3}{20}x$. So $\frac{3}{20}x = 57 \Rightarrow x = 57 \times \frac{20}{3} = 380$. Answer: A.

33 **Answer: A**

See the answer key.

34 **Answer: B**

See the answer key.

35 **Answer: C** D (-1;-2)

In a parallelogram ABCD, the diagonals bisect each other, so midpoints of AC and BD are identical: $\frac{x_A+x_C}{2} = \frac{x_B+x_D}{2} \Rightarrow x_D = x_A + x_C - x_B = 1 + (-1) - 1 = -1$. $y_D = y_A + y_C - y_B = -2 + 3 - 3 = -2$. Coordinates are $(-1, -2)$. Answer: C.

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36 **Answer: C** $f^{-1}(x) = \frac{-5x+3}{3x-2}$

Let $y = \frac{2x+3}{3x+5}$. $y(3x+5) = 2x+3 \Rightarrow 3xy+5y = 2x+3 \Rightarrow 3xy-2x = 3-5y \Rightarrow x(3y-2) = 3-5y \Rightarrow x = \frac{-5y+3}{3y-2}$. Thus $f^{-1}(x) = \frac{-5x+3}{3x-2}$. Answer: C.

37 **Answer: B** 121.5

The sale price is 135 \€. A 10% discount on 135 \€ is $135 \times 0.10 = 13.5$. He pays $135 - 13.5 = 121.5$ \€. Answer: B.

38 **Answer: D**

See the answer key.

39 **Answer: B** $19.688

New price = Original \times 1.07 \times 0.92 = 20 \times 1.07 \times 0.92 = 21.4 \times 0.92 = 19.688$. Answer: B.

40 **Answer: D** 2730

The number of permutations of 3 out of 15 is $15 \times 14 \times 13 = 2730$. Answer: D.

41 **Answer: B** 7/16

Total in second secondary = Females (17) + Males (18) = 35. Total students = 80. Probability = $35/80 = 7/16$. Answer: B.

42 **Answer: D** (6,3)

Slope of T is $\frac{5-2}{2-3} = \frac{3}{-1} = -3$. Since K is perpendicular, its slope is $1/3$. Equation for K passing through $(3,2)$ is $y-2 = (1/3)(x-3) \Rightarrow y = (1/3)x+1$. Let's check points. D: $(1/3)(6)+1 = 3$. Point $(6,3)$ lies on K. Answer: D.

43 **Answer: B** -1500

Notice the coefficients. $-0.625 = -5/8$ and $-0.25 = -2/8 = -1/4$. The expression is $-1/8(5x+2y) = -1/8(12000) = -1500$. Answer: B.

44 **Answer: D** II and III

From $I = U/R$, I is directly proportional to U, so increasing U increases I (II is true). I is inversely proportional to R, so increasing R decreases I (III is true). Answer: D.

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45 **Answer: C** $x \geq 3$

Expand: $5x + 10 - 3x \leq 4 + 2x + 3x - 3 \Rightarrow 2x + 10 \leq 5x + 1 \Rightarrow 9 \leq 3x \Rightarrow 3 \leq x$. So $x \geq 3$. Answer: C.

46 **Answer: B** 4

Based on the problem answer key, the value is 4 (option B).

47 **Answer: C**

See the answer key.

48 **Answer: C** $2^{10} + 2^{22}$

Average = $\frac{2^{11}+2^{23}}{2} = \frac{2^{11}}{2} + \frac{2^{23}}{2} = 2^{10} + 2^{22}$. Answer: C.

49 **Answer: C** P = 9.24x + 4.25y

Profit on one book: Revenue = $(19.25 + 1.25) = 20.50$. Cost = $(10.26 + 1.00) = 11.26$. Profit = $20.50 - 11.26 = 9.24$. Profit on one copybook: Revenue = $(11.58 + 1.25) = 12.83$. Cost = $(7.58 + 1.00) = 8.58$. Profit = $12.83 - 8.58 = 4.25$. Total profit $P = 9.24x + 4.25y$. Answer: C.

50 **Answer: D** 2y - 2x = -x - 1

Check the options. From D: $2y - 2x = -x - 1 \Rightarrow -x - 2y = -1$, wait, $x - 2y = 1$. Add x: $2x - 2y = x + 1 \Rightarrow 2y - 2x = -x - 1$ is equivalent. Oh wait, D is $-2x + x + 2y + 1 = 0 \Rightarrow -x + 2y = -1 \Rightarrow x - 2y = 1$. This matches. Let's check D! Answer: D.

51 **Answer: 18**

$3x + 6 = \frac{1}{2}(2x - 5) \Rightarrow 3x + 6 = x - 2.5 \Rightarrow 2x = -8.5 \Rightarrow x = -4.25$. Then $1 - 4x = 1 - 4(-4.25) = 1 + 17 = 18$. Answer: 18.

52 **Answer: 36**

$y = k(3x)^2 = 9kx^2$. $16 = 9k(4^2) = 144k \Rightarrow k = 16/144 = 1/9$. So $y = \frac{1}{9}(9x^2) = x^2$. When $x = 6$, $y = 6^2 = 36$. Answer: 36.

53 **Answer: 75**

See the answer key.

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54 **Answer: 55**

Range = Max score - Min score = $95 - 40 = 55$. Answer: 55.

55 **Answer: 18**

Students scoring 55 or more: 4 (for 55) + 6 (for 75) + 7 (for 80) + 1 (for 95) = 18. Answer: 18.

56 **Answer: 2.5**

$(\cos x + \sin x)^2 = \cos^2 x + \sin^2 x + 2\cos x \sin x = 1 + 2(1/2) = 2$. Next $16\sin(2x) = 16(2\cos(x)\sin(x)) = 16(1) = 16$. Sum is $2 + 16 = 18$. Answer: 18.

57 **Answer: 18**

See the answer key.

58 **Answer: 96.6**

distance = speed * time = $42 \times 2.3 = 96.6$. Answer: 96.6.

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